

UFUG 2106 : Discrete Mathematics

Chapter 05: Sequences, Mathematical Induction, and Recursion

Recommended readings: Pages 258-270; 277-283; 301-303; 308-309; 325-331; 315-319; 335-336; 340-348; 353-358; 365-368; 370-372

ACK: Part of the PPTs are built from the PPT slides from Instructor's Companion Website of the textbook.

Outline

1. Sequences
2. Mathematical Induction
3. Strong Mathematical Induction and the Well-Ordering Principle for the Integers
4. Recursions
5. Solving Recurrence Relations by Iteration
6. Second-Order Linear Homogeneous Recurrence Relations with Constant Coefficients
7. General Recursive Definitions and Structural Induction

Intended Learning Outcomes

By the end of this chapter, students should be able to:

1. Understand and Apply Sequences:

- Define sequences formally, recognize common types, and use summation/product notation.

2. Demonstrate Proofs via Mathematical Induction:

- Apply standard and strong mathematical induction to prove formulas and statements, as well as the Loop Invariant Theorem to establish algorithm correctness.

3. Analyze Recursive Definitions and Recurrence Relations:

- Define sequences recursively, solve simple recurrence relations using iteration, and recognize second-order linear homogeneous recurrence relations.

4. Understand and Apply General Recursive Definitions and Structural Induction:

- Extend mathematical induction to recursively defined structures.

Sequences

Definition

A **sequence** is a function whose domain is either all the integers between two given integers or all the integers greater than or equal to a given integer.

We typically represent a sequence as a set of elements written in a row. In the sequence denoted

$$a_m, a_{m+1}, a_{m+2}, \dots, a_n$$

each individual element a_k (read “a sub k”) is called a **term**.

k in a_k is called a **subscript** or **index**,

m is the subscript of the **initial term**,

n is the subscript of the **final term**.

Sequences

Definition

A **sequence** is a function whose domain is either all the integers between two given integers or all the integers greater than or equal to a given integer.

The notation

$$a_m, a_{m+1}, a_{m+2}, \dots$$

denotes an **infinite sequence**.

Sequences

An **explicit formula** or **general formula** for a sequence is a rule that shows how the values of a_k depend on k .

Define sequences $a_1, a_2, a_3, \dots$ and $b_2, b_3, b_4, \dots$ by the following explicit formulas:

$$a_k = \frac{k}{k+1} \quad \text{for every integer } k \geq 1,$$

$$b_i = \frac{i-1}{i} \quad \text{for every integer } i \geq 2.$$

Compute the first five terms of both sequences.

The Summation Notation

Definition

If m and n are integers and $m \leq n$, the symbol $\sum_{k=m}^n a_k$, read the **summation from k equals m to n of a -sub- k** , is the sum of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$. We say that $a_m + a_{m+1} + a_{m+2} + \dots + a_n$ is the **expanded form** of the sum, and we write

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots + a_n.$$

We call k the **index** of the summation, m the **lower limit** of the summation, and n the **upper limit** of the summation.

Let $a_1 = -2$, $a_2 = -1$, $a_3 = 0$, $a_4 = 1$, and $a_5 = 2$. Compute the following:

a. $\sum_{k=1}^5 a_k$

b. $\sum_{k=2}^2 a_k$

c. $\sum_{k=1}^2 a_{2k}$

The Product Notation

Definition

If m and n are integers and $m \leq n$, the symbol $\prod_{k=m}^n a_k$, read the **product from k equals m to n of a -sub- k** , is the product of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$.

We write

$$\prod_{k=m}^n a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdots a_n.$$

Compute the following products:

a. $\prod_{k=1}^5 k$

b. $\prod_{k=1}^1 \frac{k}{k+1}$

Properties of Summations and Products

Theorem 5.1.1

If $a_m, a_{m+1}, a_{m+2}, \dots$ and $b_m, b_{m+1}, b_{m+2}, \dots$ are sequences of real numbers and c is any real number, then the following equations hold for any integer $n \geq m$:

1. $\sum_{k=m}^n a_k + \sum_{k=m}^n b_k = \sum_{k=m}^n (a_k + b_k)$
2. $c \cdot \sum_{k=m}^n a_k = \sum_{k=m}^n c \cdot a_k$ generalized distributive law
3. $\left(\prod_{k=m}^n a_k \right) \cdot \left(\prod_{k=m}^n b_k \right) = \prod_{k=m}^n (a_k \cdot b_k).$

Properties of Summations and Products

Let $a_k = k + 1$ and $b_k = k - 1$ for every integer k . Write the following expression as a single summation:

$$\text{a. } \sum_{k=m}^n a_k + 2 \cdot \sum_{k=m}^n b_k$$

$$\text{a. } \sum_{k=m}^n a_k + 2 \cdot \sum_{k=m}^n b_k = \sum_{k=m}^n (k + 1) + 2 \cdot \sum_{k=m}^n (k - 1)$$

by substitution

$$= \sum_{k=m}^n (k + 1) + \sum_{k=m}^n 2 \cdot (k - 1)$$

by Theorem 5.1.1 (2)

$$= \sum_{k=m}^n ((k + 1) + 2 \cdot (k - 1))$$

by Theorem 5.1.1 (1)

$$= \sum_{k=m}^n (3k - 1)$$

by algebraic
simplification

Factorial

Definition

For each positive integer n , the quantity **n factorial** denoted $n!$, is defined to be the product of all the integers from 1 to n :

$$n! = n \cdot (n-1) \cdots 3 \cdot 2 \cdot 1.$$

Zero factorial, denoted $0!$, is defined to be 1:

$$0! = 1.$$

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \cdot 1 = 2$$

$$3! = 3 \cdot 2 \cdot 1 = 6$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5,040$$

$$8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 40,320$$

$$= 362,880$$

“n Choose r” Notation

Definition

Let n and r be integers with $0 \leq r \leq n$. The symbol

$$\binom{n}{r}$$

is read “ **n choose r** ” and represents the number of subsets of size r that can be chosen from a set with n elements.

Formula for Computing $\binom{n}{r}$

For all integers n and r with $0 \leq r \leq n$,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

These are denoted in various ways such as

$$nC_r, C(n, r), {}^nC_r, \text{ and } C_{n,r}.$$

Sometimes they are referred to as *binomial coefficients*.

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Mathematical Induction

In general, *mathematical induction* is a proof technique used to establish that a given statement holds for all natural numbers (or integers starting from some base case).

Deduction and Induction, and Mathematical Induction?

Mathematical Induction

Principle of Mathematical Induction

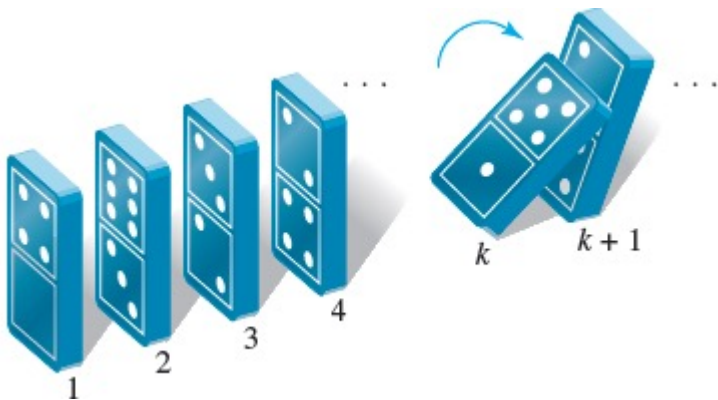
Let $P(n)$ be a property that is defined for integers n , and let a be a fixed integer. Suppose the following two statements are true:

1. $P(a)$ is true.
2. For every integer $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true.

Then the statement

for every integer $n \geq a$, $P(n)$

is true.



If the k th domino falls backward, it pushes the $(k + 1)$ st domino backward also.

1. $P(1)$ is true (the first domino falls backward)
2. For each $k \geq 1$, if $P(k)$ is true (the k th domino falls backward), then $P(k + 1)$ is also true (the $(k + 1)$ st domino falls backward).

Thus by the principle of mathematical induction, $P(n)$ (the n th domino falls backward) is true for every integer $n \geq 1$.

Method of Proof by Mathematical Induction

Method of Proof by Mathematical Induction

Consider a statement of the form, “For every integer $n \geq a$, a property $P(n)$ is true.”
To prove such a statement, perform the following two steps:

Step 1 (basis step): Show that $P(a)$ is true.

Step 2 (inductive step): Show that for every integer $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true. To perform this step,

suppose that $P(k)$ is true, where k is any particular but arbitrarily chosen integer with $k \geq a$.

*[This supposition is called the **inductive hypothesis**.]*

Then

show that $P(k+1)$ is true.

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every integer } n \geq 1.$$

Arithmetic Sequence

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for every integer } n \geq 1.$$

Arithmetic Sequence

To construct a proof by induction, you must first identify the property $P(n)$. In this case, $P(n)$ is the equation

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}. \quad \leftarrow \text{the property } (P(n))$$

Note The property is just the equation. The proof will show that the equation is true for every integer $n \geq 1$. *[To see that $P(n)$ is a sentence, note that its subject is “the sum of the integers from 1 to n ” and its verb is “equals.”]*

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every integer } n \geq 1.$$

Arithmetic Sequence

In the basis step of the proof, you must show that the property is true for $n = 1$, or, in other words, that $P(1)$ is true. Now $P(1)$ is obtained by substituting 1 in place of n in $P(n)$.

The left-hand side of $P(1)$ is the sum of all the successive integers starting at 1 and ending at 1. This is just 1.

Note To write $P(1)$, just copy $P(n)$ and replace each n by 1.

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for every integer } n \geq 1.$$

Arithmetic Sequence

Thus $P(1)$ is

$$1 = \frac{1(1+1)}{2}.$$

← basis ($P(1)$)

Of course, this equation is true because the right-hand side is

$$\frac{1(1+1)}{2} = \frac{1 \cdot 2}{2} = 1,$$

which equals the left-hand side.

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for every integer } n \geq 1.$$

Arithmetic Sequence

In the inductive step, you assume that $P(k)$ is true, for a particular but arbitrarily chosen integer k with $k \geq 1$. *[This assumption is the inductive hypothesis.]* You must then show that $P(k+1)$ is true. What are $P(k)$ and $P(k+1)$? $P(k)$ is obtained by substituting k for every n in $P(n)$.

Thus $P(k)$ is

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

← inductive hypothesis ($P(k)$)

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every integer } n \geq 1.$$

Arithmetic Sequence

Similarly, $P(k+1)$ is obtained by substituting the quantity $(k+1)$ for every n that appears in $P(n)$. Thus $P(k+1)$ is

$$1 + 2 + \cdots + (k+1) = \frac{(k+1)((k+1)+1)}{2},$$

or, equivalently,

$$1 + 2 + \cdots + (k+1) = \frac{(k+1)(k+2)}{2}. \quad \leftarrow \text{to show } (P(k+1))$$

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every integer } n \geq 1.$$

Arithmetic Sequence

Now the inductive hypothesis is the supposition that $P(k)$ is true. How can this supposition be used to show that $P(k+1)$ is true? $P(k+1)$ is an equation, and the truth of an equation can be shown in a variety of ways.

One of the most straightforward is to use the inductive hypothesis along with algebra and other known facts to separately transform the left-hand and right-hand sides until you see that they are the same.

Proof by Induction: *Sum of the First n Integers*

for every integer $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$

Arithmetic Sequence

In this case, the left-hand side of $P(k+1)$ is

$$1 + 2 + \dots + (k+1),$$

which equals

$$(1 + 2 + \dots + k) + (k+1)$$

The next-to-last term is k because the terms are successive integers and the last term is $k+1$.

By substitution from the inductive hypothesis,

$$(1 + 2 + \dots + k) + (k+1)$$

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for every integer } n \geq 1.$$

Arithmetic Sequence

$$= \frac{k(k+1)}{2} + (k+1)$$

since the inductive hypothesis says that
 $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

by multiplying the numerator and
denominator of the second term by 2 to
obtain a common denominator

$$= \frac{k^2 + k}{2} + \frac{2k + 2}{2}$$

by multiplying out the two numerators

$$= \frac{k^2 + 3k + 2}{2}$$

by adding fractions with the same
denominator and combining like terms.

Proof by Induction: *Sum of the First n Integers*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every integer } n \geq 1.$$

Arithmetic Sequence

So the left-hand side of $P(k+1)$ is $\frac{k^2 + 3k + 2}{2}$.

Now the right-hand side of $P(k+1)$ is $\frac{(k+1)(k+2)}{2} = \frac{k^2 + 3k + 2}{2}$
by multiplying out the numerator.

Thus the two sides of $P(k+1)$ are equal to each other, and so the equation $P(k+1)$ is true.

Proof by Induction: *Sum of the First n Integers*

Theorem 5.2.1 Sum of the First n Integers

For every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Definition

If a sum with a variable number of terms is shown to equal an expression that does not contain either an ellipsis or a summation symbol, we say that the sum is written **in closed form**.

Proof by Induction: *Sum of the Geometric Sequence*

Prove that $\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$, for every integer $n \geq 0$ and every real number r except 1.

Theorem 5.2.2 Sum of a Geometric Sequence

For any real number r except 1, and any integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}.$$

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Strong Mathematical Induction

- Strong mathematical induction is similar to ordinary mathematical induction in that it is a technique for establishing the truth of a sequence of statements about integers.
- Also, a proof by strong mathematical induction consists of a basis step and an inductive step.
- However, the basis step may contain proofs for several initial values, and in the inductive step the truth of the predicate $P(n)$ is assumed not just for one value of n but for all values through k , and then the truth of $P(k + 1)$ is proved.

Strong Mathematical Induction

Principle of Strong Mathematical Induction

Let $P(n)$ be a property that is defined for integers n , and let a and b be fixed integers with $a \leq b$. Suppose the following two statements are true:

1. $P(a), P(a + 1), \dots$, and $P(b)$ are all true. (**basis step**)
2. For every integer $k \geq b$, if $P(i)$ is true for each integer i from a through k , then $P(k + 1)$ is true. (**inductive step**)

Then the statement

for every integer $n \geq a$, $P(n)$

is true. (The supposition that $P(i)$ is true for each integer i from a through k is called the **inductive hypothesis**. Another way to state the inductive hypothesis is to say that $P(a), P(a + 1), \dots, P(k)$ are all true.)

Applying Strong Mathematical Induction

Prove that any integer greater than 1 is divisible by a prime number (the divisibility-by-a-prime theorem) .

Proof (by strong mathematical induction):

Let the property $P(n)$ be the sentence

n is divisible by a prime number. $\leftarrow P(n)$

Show that $P(2)$ is true:

To establish $P(2)$, we must show that

2 is divisible by a prime number. $\leftarrow P(2)$

But this is true because 2 is divisible by 2 and 2 is a prime number.

Applying Strong Mathematical Induction

Prove that any integer greater than 1 is divisible by a prime number (the divisibility-by-a-prime theorem) .

Show that for every integer $k \geq 2$, if $P(i)$ is true for each integer from 2 through k , then $P(k + 1)$ is also true:

Let k be any integer with $k \geq 2$ and suppose that

i is divisible by a prime number for each integer

i from 2 through k .

← inductive hypothesis

We must show that

$k + 1$ is divisible by a prime number.

← $P(k + 1)$

Applying Strong Mathematical Induction

Prove that any integer greater than 1 is divisible by a prime number (the divisibility-by-a-prime theorem) .

Case 1 ($k + 1$ is prime): In this case $k + 1$ is divisible by a prime number, namely, itself.

Case 2 ($k + 1$ is not prime): In this case $k + 1 = ab$ where a and b are integers with $1 < a < k + 1$ and $1 < b < k + 1$. Thus, in particular, $2 \leq a \leq k$, and so by inductive hypothesis, a is divisible by a prime number p .

In addition because $k + 1 = ab$, we have that $k + 1$ is divisible by a . Hence, since $k + 1$ is divisible by a and a is divisible by p , by transitivity of divisibility, $k + 1$ is divisible by the prime number p .

Applying Strong Mathematical Induction

Prove that any integer greater than 1 is divisible by a prime number (the divisibility-by-a-prime theorem) .

Therefore, regardless of whether $k + 1$ is prime or not, it is divisible by a prime number.

Applying Strong Mathematical Induction

Define a sequence $s_0, s_1, s_2, \dots$ as follows:

$$s_0 = 0, s_1 = 4, s_k = 6a_{k-1} - 5a_{k-2} \quad \text{for every integer } k \geq 2.$$

Prove that $s_n = 5^n - 1$.

Proof:

Let $s_0, s_1, s_2, \dots$ be the sequence defined by specifying that $s_0 = 0$, $s_1 = 4$, and $s_k = 6a_{k-1} - 5a_{k-2}$ for every integer $k \geq 2$, and let the property $P(n)$ be the formula

$$s_n = 5^n - 1. \quad \leftarrow P(n)$$

We will use strong mathematical induction to prove that for every integer $n \geq 0$, $P(n)$ is true.

Applying Strong Mathematical Induction

Define a sequence $s_0, s_1, s_2, \dots$ as follows:

$$s_0 = 0, s_1 = 4, s_k = 6a_{k-1} - 5a_{k-2} \quad \text{for every integer } k \geq 2.$$

Prove that $s_n = 5^n - 1$.

Show that $P(0)$ and $P(1)$ are true:

To establish $P(0)$ and $P(1)$, we must show that

$$s_0 = 5^0 - 1 \text{ and } s_1 = 5^1 - 1. \quad \leftarrow P(0) \text{ and } P(1)$$

But, by definition of $s_0, s_1, s_2, \dots$, we have that $s_0 = 0$ and $s_1 = 4$.

Since $5^0 - 1 = 1 - 1 = 0$ and $5^1 - 1 = 5 - 1 = 4$, the values of s_0 and s_1 agree with the values given by the formula.¹

Applying Strong Mathematical Induction

Define a sequence $s_0, s_1, s_2, \dots$ as follows:

$$s_0 = 0, s_1 = 4, s_k = 6a_{k-1} - 5a_{k-2} \quad \text{for every integer } k \geq 2.$$

Prove that $s_n = 5^n - 1$.

Show that for every integer $k \geq 1$, if $P(i)$ is true for each integer i from 0 through k , then $P(k + 1)$ is also true:

Let k be any integer with $k \geq 1$ and suppose that

$$s_i = 5^i - 1 \text{ for each integer } i \text{ with } 0 \leq i \leq k. \quad \leftarrow \text{inductive hypothesis}$$

$$\text{We must show that } s_{k+1} = 5^{k+1} - 1. \quad \leftarrow P(k + 1)$$

Applying Strong Mathematical Induction

Define a sequence $s_0, s_1, s_2, \dots$ as follows:

$$s_0 = 0, s_1 = 4, s_k = 6a_{k-1} - 5a_{k-2} \quad \text{for every integer} \quad k \geq 2.$$

Prove that $s_n = 5^n - 1$.

But since $k \geq 1$, we have that $k + 1 \geq 2$, and so

$s_{k+1} = 6s_k - 5s_{k-1}$	by definition of $s_0, s_1, s_2, \dots$
$= 6(5^k - 1) - 5(5^{k-1} - 1)$	by definition hypothesis
$= 6 \cdot 5^k - 6 - 5^k + 5$	by multiplying out and applying a law of exponents
$= (6 - 1)5^k - 1$	by factoring out 6 and arithmetic
$= 5 \cdot 5^k - 1$	by arithmetic
$= 5^{k+1} - 1$	by applying a law of exponents,

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The Well-Ordering Principle for the Integers

Well-Ordering Principle for the Integers

Let S be a set of integers containing one or more integers all of which are greater than some fixed integer. Then S has a least element.

In each case, if the set has a least element, state what it is. If not, explain why the well-ordering principle is not violated.

a. The set of all positive real numbers

b. The set of all nonnegative integers n such that $n^2 < n$

The Well-Ordering Principle for the Integers

Well-Ordering Principle for the Integers

Let S be a set of integers containing one or more integers all of which are greater than some fixed integer. Then S has a least element.

In each case, if the set has a least element, state what it is. If not, explain why the well-ordering principle is not violated.

a. The set of all positive real numbers

There is no least positive real number. For if x is any positive real number, then $x/2$ is a positive real number that is less than x . No violation of the well-ordering principle occurs because the well-ordering principle refers only to sets of integers, and this set is not a set of integers.

The Well-Ordering Principle for the Integers

Well-Ordering Principle for the Integers

Let S be a set of integers containing one or more integers all of which are greater than some fixed integer. Then S has a least element.

In each case, if the set has a least element, state what it is. If not, explain why the well-ordering principle is not violated.

b. The set of all nonnegative integers n such that $n^2 < n$

There is no *least* nonnegative integer n such that $n^2 < n$ because there is *no* nonnegative integer that satisfies this inequality. The well-ordering principle is not violated because the well-ordering principle refers only to sets that contain at least one element.

Application: Correctness of Algorithms

A program is correct if it produces the output specified in its accompanying **documentation** for each set of input data of the type specified in the documentation.

Algorithm Pre-Conditions and Post-Conditions

An **assertion** is a statement which one claims to be true.

```
python Copy  
  
def factorial(n):  
    assert n >= 0, "n must be non-negative"  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)
```

Often the predicate describing the initial state is called the **pre-condition for the algorithm**, and the predicate describing the final state is called the **post-condition for the algorithm**.

Algorithm Pre-Conditions and Post-Conditions

Algorithm to sort a one-dimensional array of real numbers

Pre-condition: The input variable $A[1], A[2], \dots, A[n]$ is a one-dimensional array of real numbers.

Post-condition: The output variable $B[1], B[2], \dots, B[n]$ is a one-dimensional array of real numbers with same elements as $A[1], A[2], \dots, A[n]$ but with the property that $B[i] \leq B[j]$ whenever $i \leq j$.

Loop Invariants

The assertion just preceding the loop is called the **pre-condition for the loop** and the one just following is called the **post-condition for the loop**. The annotated loop has the following appearance:

[Pre-condition for the loop]

while (G)

*[Statements in the body of the loop. None contain branching
statements
that lead outside the loop.]*

end while

[Post-condition for the loop]

Loop Invariants

Definition

A loop is defined as **correct with respect to its pre- and post-conditions** if, and only if, whenever the algorithm variables satisfy the pre-condition for the loop and the loop terminates after a finite number of steps, the algorithm variables satisfy the post-condition for the loop.

A **loop invariant** is a predicate with domain a set of integers, which satisfies the following condition: For each iteration of the loop, if the predicate is true before the iteration, then it is true after the iteration.

Loop Invariants

Theorem 5.5.1 Loop Invariant Theorem

Let a **while** loop with guard G be given, together with pre- and post-conditions that are predicates in the algorithm variables. Also let a predicate $I(n)$, called the **loop invariant**, be given. If the following four properties are true, then the loop is correct with respect to its pre- and post-conditions.

- I. Basis Property:** The pre-condition for the loop implies that $I(0)$ is true before the first iteration of the loop.
- II. Inductive Property:** For every integer $k \geq 0$, if the guard G and the loop invariant $I(k)$ are both true before an iteration of the loop, then $I(k + 1)$ is true after an iteration of the loop.
- III. Eventual Falsity of Guard:** After a finite number of iterations of the loop, the guard G becomes false.
- IV. Correctness of the Post-Condition:** If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.

Correctness of a Loop to Compute a Product

*[Pre-condition: m is a nonnegative integer,
 x is a real number, $i = 0$, and $product = 0$.]*

while ($i \neq m$)

1. $product := product + x$

2. $i := i + 1$

end while

[Post-condition: $product = mx$]

Use the loop invariant theorem to prove that the **while** loop is correct with respect to the given pre- and post-conditions.

Outline

1. Sequences
2. Mathematical Induction
3. Strong Mathematical Induction and the Well-Ordering Principle for the Integers
4. Recursions
5. Solving Recurrence Relations by Iteration
6. Second-Order Linear Homogeneous Recurrence Relations with Constant Coefficients
7. General Recursive Definitions and Structural Induction

Defining Sequences Recursively

A sequence can be defined in a variety of different ways.

1. Write the first few terms with the expectation that the general pattern will be obvious.
 - “consider the sequence 3, 5, 7,”
2. Define a sequence by giving an explicit formula for its i -th term.
 - a sequence $a_0, a_1, a_2 \dots$ can be specified by writing

$$a_n = \frac{(-1)^n}{n+1} \quad \text{for every integer } n \geq 0.$$

The third way to define a sequence is to use recursion.

Defining Sequences Recursively

Definition

A **recurrence relation** for a sequence $a_0, a_1, a_2, \dots$ is a formula that relates each term a_k to certain of its predecessors $a_{k-1}, a_{k-2}, \dots, a_{k-i}$, where i is an integer with $k - i \geq 0$. If i is a fixed integer, the **initial conditions** for such a recurrence relation specify the values of $a_0, a_1, a_2, \dots, a_{i-1}$. If i depends on k , the initial conditions specify the values of $a_0, a_1, \dots, a_m$, where m is an integer with $m \geq 0$.

Define a sequence $c_0, c_1, c_2, \dots$ recursively as follows: For every integer $k \geq 2$,

$$(1) \quad c_k = c_{k-1} + kc_{k-2} + 1$$

recurrence relation

$$(2) \quad c_0 = 1 \quad \text{and} \quad c_1 = 2$$

initial conditions

Find c_2, c_3 , and c_4 .

Showing That a Sequence Given by an Explicit Formula Satisfies a Certain Recurrence Relation

For each integer $n \geq 1$, $C_n = \frac{1}{n+1} \binom{2n}{n}$. (Catalan numbers)

Show that this sequence satisfies the recurrence relation $C_k = \frac{4k-2}{k+1} C_{k-1}$ for every integer $k \geq 2$.

Showing That a Sequence Given by an Explicit Formula Satisfies a Certain Recurrence Relation

For each integer $n \geq 1$, $C_n = \frac{1}{n+1} \binom{2n}{n}$. (Catalan numbers)

Show that this sequence satisfies the recurrence relation $C_k = \frac{4k-2}{k+1} C_{k-1}$ for every integer $k \geq 2$.

To obtain the k th and $(k-1)$ st terms of the sequence, just substitute k and $k-1$ in place of n in the explicit formula for $C_1, C_2, C_3, \dots$.

$$\begin{aligned} C_k &= \frac{1}{k+1} \binom{2k}{k} & C_{k-1} &= \frac{1}{(k-1)+1} \binom{2(k-1)}{k-1} \\ & & &= \frac{1}{k} \binom{2k-2}{k-1}. \end{aligned}$$

Showing That a Sequence Given by an Explicit Formula Satisfies a Certain Recurrence Relation

For each integer $n \geq 1$, $C_n = \frac{1}{n+1} \binom{2n}{n}$. (Catalan numbers)

Show that this sequence satisfies the recurrence relation $C_k = \frac{4k-2}{k+1} C_{k-1}$ for every integer $k \geq 2$.

Then start with the right-hand side of the recurrence relation and transform it into the left-hand side: For each integer $k \geq 2$,

$$\frac{4k-2}{k+1} C_{k-1} = \frac{4k-2}{k+1} \left[\frac{1}{k} \binom{2k-2}{k-1} \right]$$

by substituting

$$= \frac{2(2k-1)}{k+1} \cdot \frac{1}{k} \cdot \frac{(2k-2)!}{(k-1)!(2k-2-(k-1))!}$$

by the formula for n choose r

$$= \frac{1}{k+1} \cdot (2(2k-1)) \cdot \frac{(2k-2)!}{(k(k-1)!(k-1))!}$$

by rearranging the factors

Showing That a Sequence Given by an Explicit Formula Satisfies a Certain Recurrence Relation

For each integer $n \geq 1$, $C_n = \frac{1}{n+1} \binom{2n}{n}$. (Catalan numbers)

Show that this sequence satisfies the recurrence relation $C_k = \frac{4k-2}{k+1} C_{k-1}$ for every integer $k \geq 2$.

$$= \frac{1}{k+1} \cdot (2(2k-1)) \cdot \frac{1}{k!(k-1)!} \cdot (2k-2)! \cdot \frac{1}{2} \cdot \frac{1}{k} \cdot 2k$$

because $\frac{1}{2} \cdot \frac{1}{k} \cdot 2k = 1$

$$= \frac{1}{k+1} \cdot \frac{2}{2} \cdot \frac{1}{k!} \cdot \frac{1}{(k-1)!} \cdot \frac{1}{k} \cdot (2k) \cdot (2k-1) \cdot (2k-2)!$$

by rearranging the factors

$$= \frac{1}{k+1} \cdot \frac{(2k)!}{k!k!}$$

because $k(k-1)! = k!$, $\frac{2}{2} = 1$,
and $2k \cdot (2k-1) \cdot (2k-2)! = (2k)!$

$$= \frac{1}{k+1} \binom{2k}{k}$$

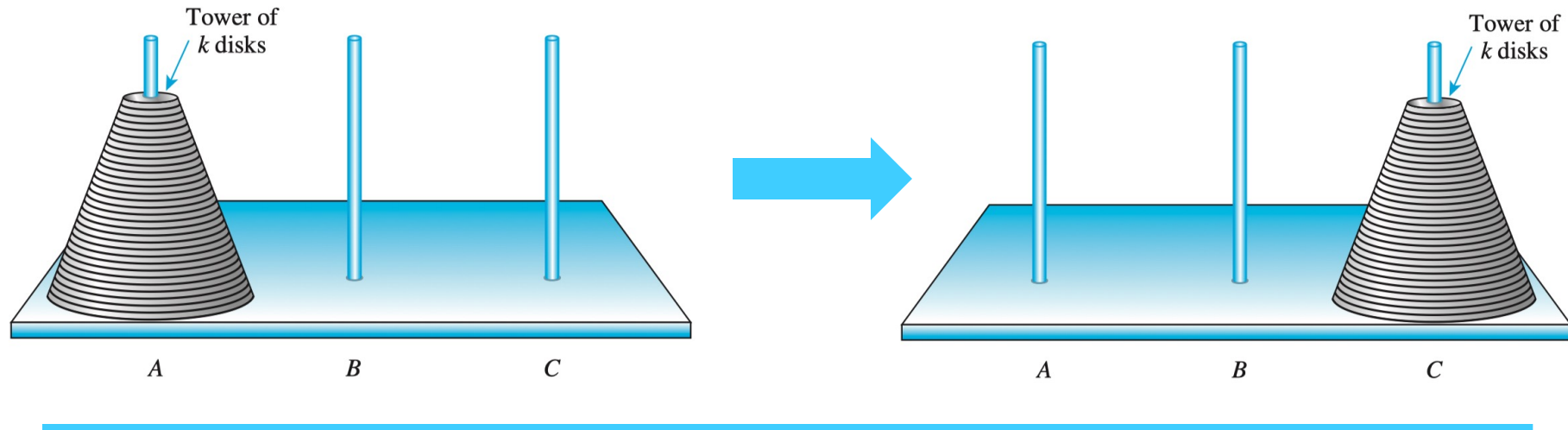
by the formula for n choose r

$$= C_k$$

by definition of $C_1, C_2, C_3, \dots$

Examples of Recursively Defined Sequences

The Tower of Hanoi

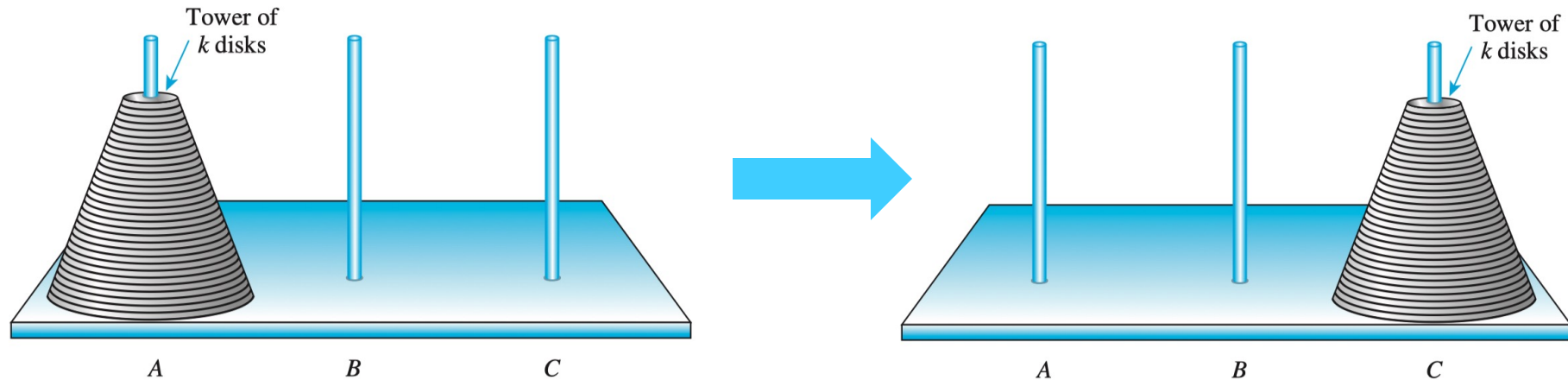


There are three towers (*A*, *B*, and *C*) and k disks of different sizes, all initially stacked on *tower A* with largest at the bottom, smallest at the top. The goal is to move all disks to *tower C*, following these rules:

1. Only one disk can be moved at a time.
2. A larger disk cannot be placed on top of a smaller disk.
3. Disks can only be moved between the three towers.

Examples of Recursively Defined Sequences

The Tower of Hanoi



Solution Approach (Recursive Method):

- **Step 1:** Move the top $k-1$ disks from tower A to tower B using tower C as an auxiliary.
- **Step 2:** Move the largest disk (the n th disk) from tower A to tower C.
- **Step 3:** Move the $k-1$ disks from tower B to tower C using tower A as an auxiliary.

$$m_k = 2m_{k-1} + 1$$

Recursive Definitions of Sum and Product

Definition

Given numbers $a_1, a_2, \dots, a_n$, where n is a positive integer, the **summation from $i = 1$ to n of the a_i** , denoted $\sum_{i=1}^n a_i$, is defined as follows:

$$\sum_{i=1}^1 a_i = a_1 \quad \text{and} \quad \sum_{i=1}^n a_i = \left(\sum_{i=1}^{n-1} a_i \right) + a_n, \quad \text{if } n > 1.$$

The **product from $i = 1$ to n of the a_i** , denoted $\prod_{i=1}^n a_i$, is defined by

$$\prod_{i=1}^1 a_i = a_1 \quad \text{and} \quad \prod_{i=1}^n a_i = \left(\prod_{i=1}^{n-1} a_i \right) \cdot a_n, \quad \text{if } n > 1.$$

A Sum of Sums

Prove that for any positive integer n , if $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

A Sum of Sums

Prove that for any positive integer n , if $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

The proof is by mathematical induction.

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i. \quad \leftarrow P(n)$$

We must show that $P(n)$ is true for every integer $n \geq 1$. We do this by mathematical induction on n .

Show that $P(1)$ is true: To establish $P(1)$, we must show that

$$\sum_{i=1}^1 (a_i + b_i) = \sum_{i=1}^1 a_i + \sum_{i=1}^1 b_i. \quad \leftarrow P(1)$$

A Sum of Sums

Prove that for any positive integer n , if $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

Now

$$\sum_{i=1}^1 (a_i + b_i) = a_1 + b_1$$

by definition of Σ

$$= \sum_{i=1}^1 a_i + \sum_{i=1}^1 b_i$$

also by definition of Σ .

Hence $P(1)$ is true.

A Sum of Sums

Prove that for any positive integer n , if $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k + 1)$ is also true:

Suppose that k is any integer with $k \geq 1$ and that $a_1, a_2, \dots, a_k, a_{k+1}$ and $b_1, b_2, \dots, b_k, b_{k+1}$ are real numbers such that

$$\sum_{i=1}^k (a_i + b_i) = \sum_{i=1}^k a_i + \sum_{i=1}^k b_i. \quad \leftarrow P(k) \text{ inductive hypothesis}$$

We must show that

$$\sum_{i=1}^{k+1} (a_i + b_i) = \sum_{i=1}^{k+1} a_i + \sum_{i=1}^{k+1} b_i. \quad \leftarrow P(k+1)$$

A Sum of Sums

Prove that for any positive integer n , if $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are real numbers, then

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.$$

Now the left-hand side of the equation is

$$\sum_{i=1}^{k+1} (a_i + b_i) = \sum_{i=1}^k (a_i + b_i) + (a_{k+1} + b_{k+1})$$

by definition of Σ

$$= \left(\sum_{i=1}^k a_i + \sum_{i=1}^k b_i \right) + (a_{k+1} + b_{k+1})$$

by inductive hypothesis

$$= \left(\sum_{i=1}^k a_i + a_{k+1} \right) + \left(\sum_{i=1}^k b_i + b_{k+1} \right)$$

by the associative and commutative laws of algebra

$$= \sum_{i=1}^{k+1} a_i + \sum_{i=1}^{k+1} b_i$$

by definition of Σ

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Solving Recurrence Relations by Iteration

Suppose you have a sequence that satisfies a certain recurrence relation and initial conditions, find an explicit formula description of the sequence.

Such an explicit formula is called a **solution** to the recurrence relation.

*The most basic method for finding an explicit formula for a recursively defined sequence is **iteration**.*

Showing That a Sequence Given by an Explicit Formula Satisfies a Certain Recurrence Relation

For each integer $n \geq 1$, $C_n = \frac{1}{n+1} \binom{2n}{n}$. (Catalan numbers)

Show that this sequence satisfies the recurrence relation $C_k = \frac{4k-2}{k+1} C_{k-1}$ for every integer $k \geq 2$.

The Method of Iteration

Given a sequence $a_0, a_1, a_2, \dots$ defined by a recurrence relation and initial conditions, the iteration works as follows:

1. Start from the initial conditions and calculate successive terms of the sequence
2. Guess an explicit formula.
3. Use mathematical induction to prove the guessed formula.

The Tower of Hanoi Sequence

Recall that the Tower of Hanoi sequence $m_1, m_2, m_3, \dots$ satisfies the recurrence relation

$$m_k = 2m_{k-1} + 1 \text{ for each integer } k \geq 2$$

and has the initial condition

$$m_1 = 1.$$

Use iteration to guess an explicit formula for this sequence.

The Tower of Hanoi Sequence

By iteration,

$$m_1 = 1$$

$$m_{\textcircled{2}} = 2m_1 + 1 = 2 \cdot 1 + 1 = 2^{\textcircled{1}} + 1,$$

$$m_{\textcircled{3}} = 2m_2 + 1 = 2(2 + 1) + 1 = 2^{\textcircled{2}} + 2 + 1,$$

$$m_{\textcircled{4}} = 2m_3 + 1 = 2(2^2 + 2 + 1) + 1 = 2^{\textcircled{3}} + 2^2 + 2 + 1,$$

$$m_{\textcircled{5}} = 2m_4 + 1 = 2(2^3 + 2^2 + 2 + 1) + 1 = 2^{\textcircled{4}} + 2^3 + 2^2 + 2 + 1.$$

The Tower of Hanoi Sequence

Thus it seems that, in general,

$$m_n = 2^{n-1} + 2^{n-2} + \cdots + 2^2 + 2 + 1.$$

By the formula for the sum of a geometric sequence (Theorem 5.2.2),

$$2^{n-1} + 2^{n-2} + \cdots + 2^2 + 2 + 1 = \frac{2^n - 1}{2 - 1} = 2^n - 1.$$

Hence the explicit formula seems to be

$$m_n = 2^n - 1 \quad \text{for every integer } n \geq 1.$$

Theorem 5.2.2 Sum of a Geometric Sequence

For any real number r except 1, and any integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}.$$

The Tower of Hanoi Sequence

Proof of Correctness: Let $m_1, m_2, m_3, \dots$ be the sequence defined by specifying that $m_1 = 1$ and $m_k = 2m_{k-1} + 1$ for each integer $k \geq 2$, and let the property $P(n)$ be the equation

$$m_n = 2^n - 1. \quad \leftarrow P(n)$$

We will use **mathematical induction** to prove that for every integer $n \geq 1$, $P(n)$ is true.

Show that $P(1)$ is true:

To establish $P(1)$, we must show that

$$m_1 = 2^1 - 1. \quad \leftarrow P(1)$$

The Tower of Hanoi Sequence

Show that $P(1)$ is true:

To establish $P(1)$, we must show that

$$m_1 = 2^1 - 1. \quad \leftarrow P(1)$$

Now the left-hand side of $P(1)$ is

$$m_1 = 1 \quad \text{by definition of } m_1, m_2, m_3, \dots,$$

and the right-hand side of $P(1)$ is

$$2^1 - 1 = 2 - 1 = 1.$$

Thus the two sides of $P(1)$ equal the same quantity, and hence $P(1)$ is true.

The Tower of Hanoi Sequence

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 1$. That is:]

Suppose that k is any integer with $k \geq 1$ such that

$$m_k = 2^k - 1. \quad \leftarrow P(k) \text{ inductive hypothesis}$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$m_{k+1} = 2^{k+1} - 1. \quad \leftarrow P(k + 1)$$

The Tower of Hanoi Sequence

But the left-hand side of $P(k + 1)$ is

$$m_{k+1} = 2m_{(k+1)-1} + 1 \quad \text{by definition of } m_1, m_2, m_3, \dots,$$

$$= 2m_k + 1$$

$$= 2(2^k - 1) + 1 \quad \text{by substitution from the inductive hypothesis}$$

$$= 2^{k+1} - 2 + 1 \quad \text{by the distributive law and the fact that } 2 \cdot 2^k = 2^{k+1}$$

$$= 2^{k+1} - 1 \quad \text{by basic algebra}$$

which equals the right-hand side of $P(k + 1)$.

The Geometric Sequence

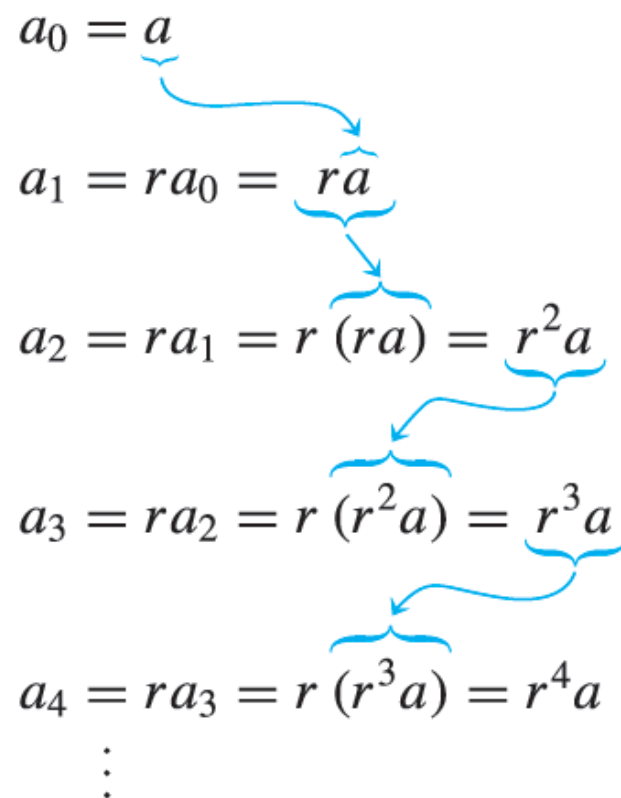
Definition

A sequence $a_0, a_1, a_2, \dots$ is called a **geometric sequence** if, and only if, there is a constant r such that

$$a_k = ra_{k-1} \quad \text{for each integer } k \geq 1.$$

Use iteration to guess an explicit formula for this sequence.

The Geometric Sequence

$$\begin{aligned} a_0 &= a \\ a_1 &= ra_0 = ra \\ a_2 &= ra_1 = r(ra) = r^2a \\ a_3 &= ra_2 = r(r^2a) = r^3a \\ a_4 &= ra_3 = r(r^3a) = r^4a \\ &\vdots \end{aligned}$$


Guess: $a_n = r^n a = ar^n$ for any arbitrary integer $n \geq 0$

The Arithmetic Sequence

Definition

A sequence $a_0, a_1, a_2, \dots$ is called an **arithmetic sequence** if, and only if, there is a constant d such that

$$a_k = a_{k-1} + d \quad \text{for each integer } k \geq 1.$$

Use iteration to guess an explicit formula for this sequence.

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Second-Order Linear Homogeneous Recurrence Relations with Constant Coefficients

Definition

A second-order linear homogeneous recurrence relation with constant coefficients is a recurrence relation of the form

$$a_k = Aa_{k-1} + Ba_{k-2} \quad \text{for every integer } k \geq \text{some fixed integer,}$$

where A and B are fixed real numbers with $B \neq 0$.

“**Second-order**” refers to the fact that the expression for a_k contains the two previous terms a_{k-1} and a_{k-2} ,

“**linear**” to the fact that a_{k-1} and a_{k-2} appear in separate terms and to the first power,

“**homogeneous**” to the fact that the total degree of each term is the same (thus there is no constant term),

“**constant coefficients**” to the fact that A and B are fixed real numbers that do not depend on k .

Second-Order Linear Homogeneous Recurrence Relations with Constant Coefficients

State whether each of the following is a second-order linear homogeneous recurrence relation with constant coefficients:

a. $a_k = 3a_{k-1} + 2a_{k-2}$

a. Yes; $A = 3$ and $B = 2$

b. $b_k = b_{k-1} + b_{k-2} + b_{k-3}$

b. No; not second-order

c. $c_k = \frac{1}{2}c_{k-1} - \frac{3}{7}c_{k-2}$

c. Yes;

d. $d_k = d_{k-1}^2 + d_{k-1} \cdot d_{k-2}$

d. No; not linear

e. $e_k = 2e_{k-2}$

e. Yes; $A = 0$ and $B = 2$

f. $f_k = 2f_{k-1} + 1$

f. No; not homogeneous

g. $g_k = g_{k-1} + g_{k-2}$

g. Yes; $A = 1$ and $B = 1$

h. $h_k = (-1)h_{k-1} + (k-1)h_{k-2}$

h. No; nonconstant coefficients

The Distinct-Roots Case

Lemma 5.8.1

Let A and B be real numbers. A recurrence relation of the form

$$a_k = Aa_{k-1} + Ba_{k-2} \quad \text{for every integer } k \geq 2 \quad 5.8.1$$

is satisfied by the sequence

$$1, t, t^2, t^3, \dots, t^n, \dots,$$

where t is a nonzero real number, if, and only if, t satisfies the equation

$$t^2 - At - B = 0. \quad 5.8.2$$

The Distinct-Roots Case

Definition

Given a second-order linear homogeneous recurrence relation with constant coefficients

$$a_k = Aa_{k-1} + Ba_{k-2} \quad \text{for every integer } k \geq 2, \quad 5.8.1$$

the characteristic equation of the relation is

$$t^2 - At - B = 0. \quad 5.8.2$$

Using the Characteristic Equation to Find Solutions to a Recurrence Relation

Consider the recurrence relation that specifies that the k th term of a sequence equals the sum of the $(k - 1)$ st term plus twice the $(k - 2)$ nd term. That is,

$$a_k = a_{k-1} + 2a_{k-2} \text{ for each integer } k \geq 2. \quad 5.8.3$$

Find all sequences that satisfy relation (5.8.3) and have the form

$$1, t, t^2, t^3, \dots, t^n, \dots,$$

where t is nonzero.

Using the Characteristic Equation to Find Solutions to a Recurrence Relation

By Lemma 5.8.1, relation (5.8.3) is satisfied by a sequence $1, t, t^2, t^3, \dots, t^n, \dots$ if, and only if, t satisfies the characteristic equation

$$t^2 - t - 2 = 0.$$

Since $t^2 - t - 2 = (t - 2)(t + 1)$,

the only possible values of t are 2 and -1 . It follows that the sequences

$$1, 2, 2^2, 2^3, \dots, 2^n, \dots \quad \text{and} \quad 1, -1, (-1)^2, (-1)^3, \dots, (-1)^n, \dots$$

are both solutions for relation (5.8.3) and there are no other solutions of this form.

Using the Characteristic Equation to Find Solutions to a Recurrence Relation

It follows that the sequences

$$1, 2, 2^2, 2^3, \dots, 2^n, \dots \quad \text{and} \quad 1, -1, (-1)^2, (-1)^3, \dots, (-1)^n, \dots$$

are both solutions for relation (5.8.3) and there are no other solutions of this form.

Note that these sequences can be rewritten more simply as

$$1, 2, 2^2, 2^3, \dots, 2^n, \dots \quad \text{and} \quad 1, -1, 1, -1, \dots, (-1)^n, \dots$$

The Distinct-Roots (相异根) Case

Lemma 5.8.2

If $r_0, r_1, r_2, \dots$ and $s_0, s_1, s_2, \dots$ are sequences that satisfy the same second-order linear homogeneous recurrence relation with constant coefficients, and if C and D are *any* numbers, then the sequence $a_0, a_1, a_2, \dots$ defined by the formula

$$a_n = Cr_n + Ds_n \quad \text{for every integer } n \geq 0$$

also satisfies the same recurrence relation.

Finding the Linear Combination That Satisfies the Initial Conditions

Consider the recurrence relation that specifies that the k th term of a sequence equals the sum of the $(k - 1)$ st term plus twice the $(k - 2)$ nd term. That is,

$$a_k = a_{k-1} + 2a_{k-2} \text{ for each integer } k \geq 2. \quad 5.8.3$$

Find a sequence that satisfies the recurrence relation (5.8.3) and that also satisfies the initial conditions

$$a_0 = 1 \text{ and } a_1 = 8.$$

Finding the Linear Combination That Satisfies the Initial Conditions

Consider the following sequences from Example 5.8.2.

$$1, 2, 2^2, 2^3, \dots, 2^n, \dots \quad \text{and} \quad 1, -1, 1, -1, \dots, (-1)^n, \dots$$

Both satisfy relation (5.8.3) although neither satisfies the given initial conditions. However, by Lemma 5.8.2, any sequence $a_0, a_1, a_2, \dots$ that satisfies the explicit formula

$$a_n = C \cdot 2^n + D(-1)^n, \tag{5.8.6}$$

where C and D are numbers, also satisfies relation (5.8.3).

Finding the Linear Combination That Satisfies the Initial Conditions

You can find C and D so that $a_0, a_1, a_2, \dots$ satisfies the initial conditions specified in this example by substituting

$n = 0$ and $n = 1$ into equation (5.8.6) and solving for C and D :

$$a_0 = 1 = C \cdot 2^0 + D(-1)^0$$

$$a_1 = 8 = C \cdot 2^1 + D(-1)^1.$$

When you simplify, you obtain the system

$$1 = C + D$$

$$8 = 2C - D,$$

which can be solved in various ways.

Finding the Linear Combination That Satisfies the Initial Conditions

For instance, if you add the two equations, you get

$$9 = 3C,$$

and so

$$C = 3.$$

Then, by substituting into $1 = C + D$, you get

$$D = -2.$$

It follows that the sequence $a_0, a_1, a_2, \dots$ given by

$$a_n = 3 \cdot 2^n + (-2)(-1)^n = 3 \cdot 2^n - 2(-1)^n,$$

for each integer $n \geq 0$, satisfies both the recurrence relation and the given initial conditions.

The Distinct-Roots Case

Theorem 5.8.3 Distinct-Roots Theorem

Suppose a sequence $a_0, a_1, a_2, \dots$ satisfies a recurrence relation

$$a_k = Aa_{k-1} + Ba_{k-2} \quad 5.8.1$$

for some real numbers A and B with $B \neq 0$ and every integer $k \geq 2$. If the characteristic equation

$$t^2 - At - B = 0 \quad 5.8.2$$

has two distinct roots r and s , then $a_0, a_1, a_2, \dots$ is given by the explicit formula

$$a_n = Cr^n + Ds^n,$$

where C and D are the numbers whose values are determined by the values a_0 and a_1 .

The Fibonacci Sequence

The Fibonacci sequence $F_0, F_1, F_2, \dots$ satisfies the recurrence relation

$$F_k = F_{k-1} + F_{k-2} \text{ for every integer } k \geq 2$$

with initial conditions

$$F_0 = F_1 = 1.$$

Find an explicit formula for this sequence.

The Fibonacci Sequence

The Fibonacci sequence satisfies the first part of the hypothesis of the distinct-roots theorem since the Fibonacci relation is a second-order linear homogeneous recurrence relation with constant coefficients ($A = 1$ and $B = 1$).

To check that it satisfies the second part of the hypothesis, examine the characteristic equation

$$t^2 - t - 1 = 0.$$

The Fibonacci Sequence

By the quadratic formula,

$$t = \frac{1 \pm \sqrt{1 - 4(-1)}}{2} = \begin{cases} \frac{1 + \sqrt{5}}{2} \\ \frac{1 - \sqrt{5}}{2} \end{cases}$$

and so the roots are distinct.

The Fibonacci Sequence

Thus it follows from the distinct-roots theorem that the Fibonacci sequence is given by the explicit formula

$$F_n = C\left(\frac{1+\sqrt{5}}{2}\right)^n + D\left(\frac{1-\sqrt{5}}{2}\right)^n \text{ for each integer } n \geq 0, \quad 5.8.7$$

where C and D are the numbers whose values are determined by the fact that $F_0 = F_1 = 1$.

The Fibonacci Sequence

To find C and D , write

$$F_0 = 1 = C\left(\frac{1+\sqrt{5}}{2}\right)^0 + D\left(\frac{1-\sqrt{5}}{2}\right)^0 = C \cdot 1 + D \cdot 1 = C + D$$

and

$$\begin{aligned} F_1 = 1 &= C\left(\frac{1+\sqrt{5}}{2}\right)^1 + D\left(\frac{1-\sqrt{5}}{2}\right)^1 \\ &= C\left(\frac{1+\sqrt{5}}{2}\right) + D\left(\frac{1-\sqrt{5}}{2}\right). \end{aligned}$$

The Fibonacci Sequence

Thus the problem is to find numbers C and D such that

$$C + D = 1$$

and

$$C\left(\frac{1+\sqrt{5}}{2}\right) + D\left(\frac{1-\sqrt{5}}{2}\right) = 1.$$

This may look complicated, but in fact it is just a system of two equations in two unknowns. By solving this two system of equations we get:

$$C = \frac{1+\sqrt{5}}{2\sqrt{5}} \quad \text{and} \quad D = \frac{-(1-\sqrt{5})}{2\sqrt{5}}.$$

The Fibonacci Sequence

Substituting these values for C and D into formula (5.8.7) gives

$$F_n = \left(\frac{1+\sqrt{5}}{2\sqrt{5}} \right) \left(\frac{1+\sqrt{5}}{2} \right)^n + \left(\frac{-(1-\sqrt{5})}{2\sqrt{5}} \right) \left(\frac{1-\sqrt{5}}{2} \right)^n,$$

or, simplifying,

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+1} \quad 5.8.8$$

for each integer $n \geq 0$. Remarkably, even though the formula for F_n involves $\sqrt{5}$, of the values of the Fibonacci sequence are integers.

The Single-Root Case

Lemma 5.8.4

Let A and B be real numbers and suppose the characteristic equation

$$t^2 - At - B = 0$$

has a single root r . Then the sequences $1, r^1, r^2, r^3, \dots, r^n, \dots$ and $0, r, 2r^2, 3r^3, \dots, nr^n, \dots$ both satisfy the recurrence relation

$$a_k = Aa_{k-1} + Ba_{k-2}$$

for each integer $k \geq 2$.

The Single-Root Case

Theorem 5.8.5 Single-Root Theorem

Suppose a sequence $a_0, a_1, a_2, \dots$ satisfies a recurrence relation

$$a_k = Aa_{k-1} + Ba_{k-2}$$

for some real numbers A and B with $B \neq 0$ and for every integer $k \geq 2$. If the characteristic equation $t^2 - At - B = 0$ has a single (real) root r , then $a_0, a_1, a_2, \dots$ is given by the explicit formula

$$a_n = Cr^n + Dnr^n,$$

where C and D are the real numbers whose values are determined by the values of a_0 and any other known value of the sequence.

The Single-Root Case

Suppose a sequence $b_0, b_1, b_2, \dots$ satisfies the recurrence relation

$$b_k = 4b_{k-1} - 4b_{k-2} \text{ for every integer } k \geq 2, \quad 5.8.11$$

with initial conditions

$$b_0 = 1 \text{ and } b_1 = 3.$$

Find an explicit formula for $b_0, b_1, b_2, \dots$

The Single-Root Case

This sequence satisfies part of the hypothesis of the single-root theorem because it satisfies a second-order linear homogeneous recurrence relation with constant coefficients ($A = 4$ and $B = -4$).

The single-root condition is also met because the characteristic equation

$$t^2 - 4t + 4 = 0$$

has the unique root $r = 2$ [since $t^2 - 4t + 4 = (t - 2)^2$].

The Single-Root Case

It follows from the single-root theorem that $b_0, b_1, b_2, \dots$ is given by the explicit formula

$$b_n = C \cdot 2^n + Dn2^n \quad \text{for each integer } n \geq 0, \quad 5.8.12$$

where C and D are the real numbers whose values are determined by the fact that $b_0 = 1$ and $b_1 = 3$. To find C and D , write

$$b_0 = 1 = C \cdot 2^0 + D \cdot 0 \cdot 2^0 = C$$

and

$$b_1 = 3 = C \cdot 2^1 + D \cdot 1 \cdot 2^1 = 2C + 2D.$$

The Single-Root Case

Hence the problem is to find numbers C and D such that

$$C = 1$$

and

$$2C + 2D = 3.$$

Substitute $C = 1$ into the second equation to obtain

$$2 + 2D = 3,$$

and so

$$D = \frac{1}{2}.$$

The Single-Root Case

Now substitute $C = 1$ and D into formula (5.8.12) to conclude that

$$b_n = 2^n + \frac{1}{2}n2^n = 2^n\left(1 + \frac{n}{2}\right) \quad \text{for each integer } n \geq 0.$$

OutlineOutline

1. Sequences
2. Mathematical Induction
3. Strong Mathematical Induction and the Well-Ordering Principle for the Integers
4. Recursions
5. Solving Recurrence Relations by Iteration
6. Second-Order Linear Homogeneous Recurrence Relations with Constant Coefficients
7. General Recursive Definitions and Structural Induction

Recursively Defined Sets

Formally, a recursive definition for a set consists of the following three components:

- I. **Base:** A statement that certain objects belong to the set.
- II. **Recursion:** A collection of rules indicating how to form new set objects from those already known to be in the set.
- III. **Restriction:** A statement that no objects belong to the set other than those coming from the base and the recursion.

Recursive Definition of Boolean Expressions

I. Base: Each symbol of the alphabet is a Boolean expression.

II. Recursion: If P and Q are Boolean expressions, then the following are also Boolean expressions:

$$\text{II(a) } P \wedge Q \quad \text{II(b) } P \vee Q \quad \text{II(c) } \sim P \quad \text{II(d) } (P)$$

III. Restriction: There are no Boolean expressions over the alphabet other than those obtained from the base and the recursion.

Derive the fact that the following is a Boolean expression over the English alphabet $\{a, b, c, \dots, x, y, z\}$:

$$\sim(p \wedge q) \vee (\sim r \wedge p).$$

Recursive Definition of Boolean Expressions

Derive the fact that the following is a Boolean expression over the English alphabet $\{a, b, c, \dots, x, y, z\}$: $\sim(p \wedge q) \vee (\sim r \wedge p)$.

(1) By I, p , q , and r are Boolean expressions.

II(a) $P \wedge Q$ II(b) $P \vee Q$ II(c) $\sim P$ II(d) (P)

(2) By (1), II(a), and II(c), $p \wedge q$ and $\sim r$ are Boolean expressions.

(3) By (2), II(d), and II(a), $(p \wedge q)$ and $\sim r \wedge p$ are Boolean expressions.

(4) By (3), II(c), and II(d), $\sim(p \wedge q)$ and $(\sim r \wedge p)$ are Boolean expressions.

(5) By (4) and II(b), $\sim(p \wedge q) \vee (\sim r \wedge p)$ is a Boolean expression.

Recursive Definition of Strings over a Finite Set

Recursive Definition for the Set of All Strings over a Finite Set

Let A be any finite set. Call the elements of A **characters**, and define the **set S of all strings over A** as follows:

- I. Base: λ is a string in S , where λ denotes the **null string**, the “string” with no characters.
- II. Recursion: New strings are formed according to the following rules:

II(a) If u is any string in S and if c is any character in A , then

uc is a string in S ,

where uc is called the **concatenation of u and c** , and is obtained by appending c on the right of u .

II(b) If u is any string in S , then both the concatenation of λ and u , denoted λu , and the concatenation of u and λ , denoted $u\lambda$, are defined to equal u . Symbolically:

$$\lambda u = u\lambda = u.$$

II(c) If u and v are any strings in S , and if c is any character in A , then the concatenation of u and vc is defined to equal the concatenation of uv and c . Symbolically:

$$u(vc) = (uv)c.$$

- III. Restriction: Nothing is a string in S other than objects obtained from the base and the recursion.

Recursive Definition of Strings over a Finite Set

Theorem 5.9.1 Characters Are Strings

If A is a finite set and S is the set of all strings over A , then every character in A is a string in S .

Proving a Property of Strings

Suppose A is a finite set, S is the set of all strings over A , and a and b are in A . Prove that $b(aa)$ is a string in S .

- (1) By Theorem 5.9.1, a and b are strings in S because both a and b are in A .
- (2) By (1) and $II(a)$, ba is a string in S because b is in S and $a \in A$.
- (3) By (2) and $II(a)$, $(ba)a$ is a string in S because $ba \in S$ and $a \in A$.
- (4) By $II(c)$, $b(aa) = (ba)a$ because b and a are strings in S and $a \in A$.
- (5) By (3) and (4), $b(aa)$ is a string in S because it equals $(ba)a$, which is a string in S .

Proving Properties about Recursively Defined Sets

Structural Induction for a Recursively Defined Set

Let S be a set that has been defined recursively, and let $P(x)$ be a property that objects in S may or may not satisfy. To prove that every object in S satisfies $P(x)$, perform the following two steps:

Step 1 (basis step): Show that $P(a)$ is true for each object a in the base for S .

Step 2 (inductive step): Show that for each x in S , if $P(x)$ is true and if y is obtained from x by applying a rule from the recursion, then $P(y)$ is true. To perform this step,

suppose that x is an arbitrarily chosen element of S
for which $P(x)$ is true.

*[This supposition is the **inductive hypothesis**.]*

Then

show that if y is obtained from x by applying a
rule from the recursion for S , then $P(y)$ is true.

Conclusion: Because no objects other than those obtained from the base and recursion are contained in S , steps 1 and 2 prove that $P(x)$ is true for every object x in S .

Proving Properties about Recursively Defined Sets

Definition Length of a String

Given the set of all strings S over a finite set A , the **length L of a string in S** is defined as follows:

1. $L(\lambda) = 0$.
2. For every string u in S and for every character a in A , the length of ua is one more than the length of u . Symbolically:

$$L(ua) = L(u) + 1 \quad \text{where} \quad u \in S \text{ and } a \in A.$$

Proving Properties about Recursively Defined Sets

Theorem 5.9.2 Additive Property of String Length

If S is the set of all strings over a finite set A , then for all strings u and v in S ,
 $L(uv) = L(u) + L(v)$.

Proving Properties about Recursively Defined Sets

Theorem 5.9.3 The Concatenation of Any Two Strings Is a String

If S is the set of all strings over a finite set A and u and v are any strings in S , then uv is a string in S .

Theorem 5.9.4 Concatenation of Strings Is Associative

If S is the set of all strings over a finite set A and u , v , and w are any strings in S , then $u(vw) = (uv)w$.

Recursive Functions

A function is said to be **defined recursively** or to be a **recursive function** if its rule of definition refers to itself.

McCarthy's 91 Function $M : \mathbf{Z}^+ \rightarrow \mathbf{Z}$

$$M(n) = \begin{cases} n - 10 & \text{if } n > 100 \\ M(M(n + 11)) & \text{if } n \leq 100 \end{cases}$$

for all positive integers n . Find $M(99)$.

A Recursive “Function” That Is Not Well Defined

Consider the following attempt to define a recursive function G from $\mathbf{Z}^+$ to $\mathbf{Z}$. For each integer $n \geq 1$,

$$G(n) = \begin{cases} 1 & \text{if } n \text{ is } 1 \\ 1 + G\left(\frac{n}{2}\right) & \text{if } n \text{ is even} \\ G(3n - 1) & \text{if } n \text{ is odd and } n > 1. \end{cases}$$

Is G well defined? Why?